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Introduction To Intellectual Property: Role of IP in the Economic and Cultural Development of the Society, IP Governance, IP as a Global Indicator of Innovation, Origin of IP History of IP in India. Major Amendments in IP Laws and Acts in India.

Patents: Conditions for Obtaining a Patent Protection, To Patent or Not to Patent an Invention. Rights Associated with Patents. Enforcement of Patent Rights. Inventions Eligible for Patenting. Non-Patentable Matters. Patent Infringements. Avoid Public Disclosure of an Invention before Patenting. Process of Patenting.

Process of Patenting: Prior Art Search. Choice of Application to be Filed. Patent Application Forms. Jurisdiction of Filing Patent Application. Publication. Pre-grant Opposition. Examination. Grant of a Patent. Validity of Patent Protection. Post-grant Opposition. Commercialization of a Patent. Need for a Patent Attorney/Agent. Can a Worldwide Patent be Obtained? Do I Need First to File a Patent in India? Patent Related Forms. Fee Structure. Types of Patent Applications. Commonly Used Terms in Patenting. National Bodies Dealing with Patent Affairs. Utility Models.

Introduction To Intellectual Property

- **Definition of Intellectual Property** : Intellectual Property (IP) is a special category of property created by human intellect in various fields such as arts, literature, science, trade, etc. Since IP is a novel creation of the mind, it is intangible (invisible and indivisible) in nature, differing from tangible properties like land, house, gold, and car.
- **Intellectual Property Rights (IPR)**: IPR refers to the privileges accorded to the creator or inventor. These rights are given in exchange for revealing the process of creation or invention to the public.
- The inventor is conferred with special rights to use, sell, distribute, offer for sale, and restrict others from using the invention without prior permission.

Introduction To Intellectual Property

- **Purpose of IPR :** The primary purposes of IPR include
 - ❖ Encouraging innovation and creativity
 - ❖ Protecting the rights of creators and inventors
 - ❖ Maintaining a balance between public access and private rights.
 - **Types of Intellectual Property :** Broadly, IP comprises two branches: Copyrights and Related Rights, and Industrial Property Rights.
- 1. Copyrights and Related Rights :** Copyrights and Related Rights refer to creative expressions such as books, publications, architecture, music, wood/stone carvings, pictures, portrays, sculptures, films, and software/databases.
 - 2. Industrial Property Rights :** Industrial Property Rights focus on protecting innovations and brands. This category includes patents, trademarks, trade services, industrial designs, and geographical indications

Introduction To Intellectual Property

- **Key Features of IP Categories:**
- **Copyrights (Creative expressions):** Copyright is the right bestowed on the owner or creator in relation to publication, and distribution of a piece of writing, music, picture or related works. Copyright also applies to technical contents such as software, datasheets and related documents.
- **Patents (Protecting inventions):** A patent is a legal record that bestows the holder the exclusive right over an invention as per the claims, in a limited geographical domain and for a limited duration by thwarting possible interested parties from any form of manufacture, use or sale of the product or outcome of the invention.
- **Trademarks (Brand identity):** A trademark is a sign that suitably differentiates the owner's goods or services from those of others.

Introduction To Intellectual Property

- **Trade services:** Any services in relation to trade or any trade related financing, lending or other financial accommodation provided(or to be provided) by the bank, including but not limited to issuance/amendment of letter of credit, document arrival under letter of credit, application for negotiation and inquiries etc.
- **Industrial Designs(Aesthetic aspects):** An industrial design protection is related to certain specific ornamental shapes associated with products whose duplication the owner may wish to prevent.
- **Geographical Indications(Origin-based products):** A geographical indication (GI) is a name or sign used on products which corresponds to a specific geographical location or origin. Items that meet geographical origin and quality standards may be endorsed with a government-issued stamp which acts as official certification of the origins and standards of the product.

Role of IP in the Economic and Cultural Development of the Society

❖ Creativity and Progress:

- Creativity being the keystone of progress, no civilized society can afford to ignore the basic requirement of encouraging the same.
- The economic and social development of a society is largely dependent on creativity.
- The protection provided by the IPR to the creators/innovators is in fact an act of incentivization (providing legal protection and exclusive rights to creators) for encouraging them to create more and motivates others to create new.

❖ Negative Impact of Rigid IPR: However, if IPR is practiced rigidly, it may have a negative impact on the progress of society.

- For example, compliance with the Trade Related Aspects of Intellectual Property Rights (TRIPS) Agreement has affected the farming community as they are unable to store seeds for the next crop.
- Multinational companies regulate the price of seeds, which is generally beyond the reach of a majority of the farmers.

Role of IP in the Economic and Cultural Development of the Society

- ❖ **Balancing IPR with Exceptions:** To address or mitigate the negative impact of IPR, certain laws, exceptions and limitations associated with IPR have been enacted to maintain a balance between the interests of the creators/inventors and the community.
- For example, farmers rights under the **Protection of Plant Varieties and Farmers Rights (PVP&FR) Act, 2001** entitles them to many privileges, such as Rights on seeds provides rights to the farmers to save seeds, use seeds and share, exchange or sell seeds to other farmers.
- Right to protection protects the farmers from infringement and other legal accusation levied upon them due to his legal ignorance in using other's plant varieties.
- The use of copyrighted material for education and religious ceremonies is exempted from the operation of the rights granted in the Copyright Act.
- Similarly, a patent can be revoked in favor of compulsory licensing by the government during an emergency or a natural calamity.

Role of IP in the Economic and Cultural Development of the Society

- **Society's Interest and Government Intervention:** If an invention/creation is not in the interest of society, it is not registered by the government for grant of any rights associated with IP.
- For example, cloning of human embryos is banned for IP protection, and so is the creation of super microbial pathogens, which can play havoc with human lives.
- **Traditional Knowledge (TK) and Genetic Resources:** India is enriched with massive biodiversity and genetic resources and their use is embodied in what is referred to as Traditional Knowledge (TK).
- However, the use of such knowledge and resources are not limited to local contexts as many innovations relate to and draw on them.
- Therefore, the main issue of concern is to protect TK and genetic resources, which are rapidly coming under the governance of sometimes conflicting IPR policies.
- To derive maximum benefit from them, the establishment of adequate legal infrastructure and enforcement is required

Role of IP in the Economic and Cultural Development of the Society

- **Initiatives and Benefits:** With initiatives like 'Make in India', 'Atmanirbhar Bharat' and supporting local homegrown brands, and easy as well as accessible approach to patents and trademarks registration, it is possible to reap the benefits of our resources.

IP Governance(Role of National Agencies and International Organizations)

- **National IP Governance in India** : Since IP is an integral component of human society, each and every nation has dedicated agencies for laying out the guidelines, implementation and enforcement of IP related matters.
- In India, many organizations/agencies deal with various aspects of IP.
- The governance of all categories of IP, except the Plant Variety and Farmers' Rights Act, is carried out by the **Department for Promotion of Industry & Internal Trade (DPIIT)** under the guidance, sponsorship, or **Authority of the Ministry of Commerce and Industry**, Government of India (GoI).

IP Governance(Role of National Agencies and International Organizations)

- **Indian Organizations for IP:** There are a few other dedicated organizations/departments established by the government to promote patent-ecosystem (patent awareness, patent filing and patent commercialization) in India e.g.
 - i. TIFAC (Technology Information Forecasting and Assessment Council)
 - ii. NRDC (National Research Development Corporation)
 - iii. CIPAM (Cell for IPR Promotion and Management)

International IP Governance - WIPO : In order to create a hassle-free exchange of IP related activities amongst all the nations, the United Nations (UN) has established an organization called the **World Intellectual Property Organization (WIPO)** to set minimum standards of rules and regulations pertaining to all aspects of IP including rights, empowerment, exceptions, etc. .

This agency is at the forefront of imparting knowledge about IP and governs international filing and registration of IP through various Conventions and Treaties like **Paris Conventions, Patent Cooperation Treaty (PCT), Rome Convention, Berne Convention**, etc.

IP as a Global Indicator of Innovation

- **IP as an Indicator of Innovation :** Intellectual Property (IP), especially patents, plays a crucial role as an indicator of innovation.
- Global ranking organizations use IP as a parameter to evaluate Science, Technology, and Innovation (STI) ecosystems.
- Example: **Scimago 2020 Report:** The Scimago 2020 Report, a publicly available online portal ranking journals and countries based on Scopus data, highlights India's position.
- In the report, India is ranked 4th position in the parameter of 'Research Publications' but 50th position in the parameter of 'Intellectual Property Rights.'

IP as a Global Indicator of Innovation

- **Importance of IP in Global Rankings** : IP is reflective of a nation's innovation and research output.
- There is a need to sensitize teaching and scientific communities about the importance of IP.
- Creating infrastructure in higher learning institutes for IP awareness and utilization is crucial.
- **Origin of IP**
 - Ancient IP Practices: Though there is no official record of the origin of IP, rudimentary forms of IP were believed to exist
 - Sybaris Example : Natives of Sybaris were granted a year's protection for their intellect to create "any new improvement in luxury".
 - Medieval Europe : Practical IP governance began taking shape in medieval Europe.

IP as a Global Indicator of Innovation

- **1623 Legislation in Britain** : Britain passed an IP legislation in 1623, allowing guilds(association of artisans or merchants) to create innovations and bring them to market for trade purposes. However, it faced public resentment and was replaced by the 'Statute of Monopolies.
- **Statute of Monopolies (1624)**: 'Statute of Monopolies' granted rights to the original creator/inventor for 14 years, addressing public concerns.
- **Statute of Anne (1710)**: Another legislation, 'Statute of Anne', was passed by the British parliament in 1710. This legislation aimed at strengthening copyrights by providing rights to the authors for recreation and distribution of their work. The work could also be renewed for another 14 years.

IP as a Global Indicator of Innovation

- **Global Adoption of IP Legislation** : By the end of the 18th century and the beginning of the 19th century, almost every country started laying down IP legislation to protect their novel inventions and creations.

History of IP in India

- History of IP covering the pre-independence era, amendments and developments in India
- History of IP covering the post-independence era
- Key milestones in shaping Indian patent laws

History of patents

- **Patents in Pre-Independence Era:**
- The first patent-related legislation in India was Act VI of 1856, adapted from the British Patent Law of 1852
- Objective: To encourage the inventions of new and useful manufactures.
- The rights conferred to the inventor were termed as 'Exclusive Privileges'.
- Provided **exclusive privileges** to inventors for 14 years.

History of patents

- **Amendments in 1859 and Developments: Act XV of 1859**
- Increase of priority time from 6 months to 12 months.
- Granted **exclusive privileges** for making, selling, and using inventions in India.
- Duration: **14 years** from the date of filing.
- Only useful inventions qualified.
- Exclusion of importers from the definition of the inventor
- The world's first patent was granted in 1790 to Samuel Hopkins in the USA for the "making of potash and pearl ash by a new apparatus and process".
- In India, the first patent (known as 'Exclusive Privileges' at that time) was awarded in 1856 to a civil engineer, George Alfred DePenning from Calcutta, for his invention, 'An Efficient Punkah Pulling Machine'.

History of patents



History of patents

- Inclusion of 'Designs' in Patent Act:
- The new Act was rechristened as "The Patterns and Designs Protection Act" under Act XIII of 1872 to include the 'Designs' into patent Act.
- This Act was further amended in 1883 (XVI of 1883) to include the provision of protection for 'Novelty' in the invention.
- The Indian Patents and Designs Act, 1911: At the beginning of the 20th century, all the earlier Acts related to inventions and designs were done away with the introduction of 'The Indian Patents and Designs Act, 1911' (Act II of 1911).
- As per this Act, the governance of patents was placed under the management of the Controller of Patents.
- In the next three decades, many amendments were introduced for reciprocal arrangements with other countries for securing priority dates.

History of patents

- These amendments dealt with:
- Use of invention by the government.
- Patent of Addition.
- Enhancing the term of the patent from 14 years to 16 years.
- Filing of 'Provisional Application' and submission of 'Complete Application' within 9 months from the date of filing the application.

History of patents

- **Patents in Post-Independence Era :**
- After India got independence in 1947, many patent experts felt the need to review the Indian Patents and Designs Act, 1911, keeping the national interest (economic and political) in mind.
- A dedicated committee, chaired by a renowned Justice Bakshi Tek Chand (retired Judge of Lahore High Court), was constituted in 1949 to review the advantages of the patent system.
- The committee submitted a plethora of recommendations, including :
- Misuse of patents rights needs to be prevented.
- There must be a clear indication in the Act that food, medicine and surgical and curative devices should be made available to the masses at the cheapest rate by giving reasonable compensation to the owner of the patent.
- Amendments in Sections 22, 23 and 23A of the Patent and Design Act, 1911, on the lines of the UK Patent Act.

History of patents

- **Patents in Post-Independence Era :**
- **Amendments in 1950 and 1952:** These recommendations were introduced in the Act XXXII of 1950.
- Two years later, another amendment (Act LXX of 1952) was made to provide compulsory licensing of patents related to food, drugs and chemicals killing insects and microbes.
- Based on these amendments, a bill was presented in the parliament in 1953 but was rejected.
- **Committee in 1957 and New Legislation in 1965:**
- In 1957, the central government constituted another powerful committee under the chairmanship of Justice N. Rajagopala Ayyangar to seek inputs for further strengthening the Indian Patent Law.
- The committee submitted its report to the government in 1959.

History of patents

- **Patents in Post-Independence Era :**
- **New Legislation in 1965:**
- It comprised two segments addressing:
 - General aspects of the patent laws.
 - Bill rejected back in 1953
- **Patents Act, 1970:**
- The revised patent legislation was submitted to the Lok Sabha in 1965. After many hiccups, clarifications, and modifications, the Patents Act, 1970 was introduced, superseding all the previous laws related to the patents. However, the Indian Patents and Designs Act of 1911 remained applicable for designs only until 1994.
- Reference:(http://www.ipindia.nic.in/writereaddata/Portal/IPOAct/1_31_1_patentact-1970-11march2015.pdf)

History of patents

- **Patents in Post-Independence Era :**
- **India Signs TRIPS Agreement in 1995:** In 1995, India signed the TRIPS Agreement and got a transition period of 10 years (1995-2005) to make domestic laws compatible with
- **Patents (Amendment) Act, 1999 :** In 1999, The Patents (Amendment) Act, 1999 was introduced providing for the filing of applications for 'Product Patents' in the areas of drugs, pharmaceuticals and agrochemicals (earlier, only 'processes' were protected under the Patent Act).
- The new Patent Act also included provisions for the grant of Exclusive Market Rights (EMRs) for the distribution and sale of pharmaproducts on fulfillment of certain conditions the international treaty.

History of patents

- Patents in Post-Independence Era :
- The **TRIPS Agreement** (Agreement on Trade-Related Aspects of Intellectual Property Rights) is a [World Trade Organization \(WTO\)](#) treaty that sets minimum standards for intellectual property (IP) protection across its member countries. It was adopted in 1994 and entered into force in 1995, establishing rules for copyrights, trademarks, patents, geographical indications, industrial designs, and trade secrets to facilitate international trade and resolve disputes related to IP

History of patents (Patents in Post-Independence Era)

- **Patents (Amendment) Act, 2002:** The second amendment to the 1970 Act was made through the Patents (Amendment) Act, 2002 (Act 38 of 2002).
- This Act introduced new Patent Rules, 2003, thus replacing the earlier Patents Rules, 1972.
- **The major amendments were:**
 - The protection term of 20 years for all inventions from the date of filing.
 - Scope of non-patentable inventions including Traditional Knowledge expanded.
 - Disclosure of source and geographical origin of biological material made compulsory.
 - Provisions concerning convention countries simplified.
 - Establishment of Appellate Board.
 - Compulsory license provisions strengthened.
 - Simplification of procedures.
 - Harmonization with Patent Cooperation Treaty (PCT) provisions.

History of patents (Patents in Post-Independence Era)

- The Intellectual Property Appellate Board, a quasi-judicial body, was constituted in September 1958. The jurisdiction of the Copyright Board extends to the whole of India. The Board is entrusted with the task of adjudication of disputes pertaining to copyright registration, assignment of copyright, grant of Licenses in respect of works withheld from public, unpublished Indian works, production and publication of translations and works for certain specified purposes. It also hears cases in other miscellaneous matters instituted before it under the Copyright Act, 1957.
- The Copyright(Amendment)Act, 2012 which came into force w.e.f. June 21,2015 provides for a full time IPAB Under Section 11 with Chairman and two other memnbers,with its headquarters in Delhi.The new full time IPAB is expected to be reconstituted shortly.

History of patents (Patents in Post-Independence Era)

- **Patents (Amendment) Act, 2005** : With the rapidly changing scenario of IPR at a global level, a need was felt to further amend the Patent Act, 1970. The highlight of the
- Patents (Amendments) Act 2005 were
- Product patent for inventions in all fields of technology.
- New forms of known substances excluded to prevent evergreening of the patent.
- Rationalization of the opposition procedure.
- Introduction of pregrant opposition by representation.
- Introduction of post-grant opposition.
- Compulsory license for export purposes.
- Compulsory license for manufacture.
- Extension of grace period from 6 months to 12 months for filing a patent, if published in government exhibition.

History of patents (Patents in Post-Independence Era)

- **International Agreements** : India is a member of all prominent Conventions and Treaties related to the facilitation of the inventors for international filing and protecting the rights over the inventions globally.
- The important international agreements to which India is a signatory party are TRIPS Agreement (1995), Paris Convention (1883), PCT (1970) and Budapest Treaty (1977) and many more.

Patents

- A patent is an exclusive right granted for an innovation that generally provides a new way of doing something or offers a new technical solution to a problem.
- The exclusive right legally protects the invention from being copied or reproduced by others.
- In return, the invention must be disclosed in an application in a manner sufficiently clear and complete to enable it to be replicated by a person with an ordinary level of skill in the relevant field.

Conditions for Obtaining Patent Protection

- There is a set criterion, as provided in Section 2(1)(j) of the Patents Act, 1970, which must be fulfilled for a product or a process to qualify for the grant of a patent. The criterion encompasses:
- **Novelty** - Not part of 'State of the Art'. The innovation claimed in the patent application is new and not known to anybody in the world. In other words, the innovation is a) not in the knowledge of the public, b) not published anywhere through any means of publication, and c) not claimed in any other specification by any other applicant.
- **Inventive Step** - Not obvious to the person(s) skilled in the art. The innovation is a) a technical advancement over the existing knowledge, b) possesses economic significance, and c) not obvious to a person skilled in the concerned subject.
- **Capable of Industrial Application** - For the benefit of society. The invention is capable of being made or used in any industry.

To Patent or Not to Patent an Invention

- Once an invention has been developed, the inventor has to decide whether to exploit the invention for personal benefits as provided by the statutory laws of the country or put it in the public domain.
- By and large, the inventor prefers the former option.
- Only an extremely small amount of inventions are placed in the public domain without claiming any benefits.
- In the latter case, anybody can exploit the innovation for commercial or societal benefit without paying any money to the inventor.
- If the owner of an invention wishes to seek monetary gains, he can choose from either of the two options, i.e. patenting or Trade Secret.
- If the inventor is absolutely sure of maintaining the secrecy of the invention for a very long period (maybe 100 years or more) and the probability of reverse engineering of the technology is nil or very low, then the 'Trade Secret' category is preferred.

- If the invention has a short life span or can be kept secret only for a small period of time (a couple of years or so) or the probability of reverse engineering is high once the invention is in the public domain, then the 'patent' category is preferred.

Rights Associated with Patents

- As per the Court of Law, a patent owner has the right to decide who may or may not use the patented invention.
- In other words, the patent protection provided by the law states that the invention cannot be commercially made, used, distributed, imported, or sold by others without the patent owner's consent.
- The patent owner may permit other parties to use the invention on mutually agreed terms.
- As a matter of fact, the patent rights are negative rights as the owner is restricting others from using the patent in any manner without his prior permission.
- The patent holder may choose to sue the infringing party to stop illegal use of the patent and also ask for compensation for the unauthorized use.

Inventions Eligible for Patenting

- **Range of Patentable Inventions:** Patents may cover inventions in diverse fields, from everyday items like paper clips to advanced technologies such as nanotechnology chips or genetically modified organisms.
- **Everyday Products with Patents:** Various everyday products, including toothbrushes, shoes, pens, eyeglasses, textiles, and electronic devices, often incorporate patented technologies.
- Complex products like laptops, cars, and mobile phones may involve numerous patented components.
- **Misconceptions about Patent Awards:** Contrary to common belief, patents are not exclusive to major scientific breakthroughs. Many patents are granted for incremental improvements over existing inventions.
- **Examples of Patentable Molecules:** Single molecules, like penicillins, and their derivatives can receive multiple patents. These derivatives, achieved through structural modifications, exhibit new and improved properties.

Inventions Excluded from Patenting

- The Patent Act, 1970 outlines certain exclusions regarding products and processes that are not eligible for patents. These exclusions include:
- **Inventions Contrary to Public Morality:** Examples: Methods for human cloning, methods for gambling.
- **Mere Discovery:** Examples: Discovery of a new micro-organism occurring freely in nature, laws of gravity.
- **Frivolous Invention:** An invention that lacks practical utility or industrial application, meaning it does not provide any real technical advancement, usefulness, or benefit to society.
- Examples: Dough supplemented with herbs merely changing the taste, a 100 years calendar, bus timetable.

Inventions Excluded from Patenting

- **Arrangement or Rearrangement: Examples:** An umbrella fitted with a fan, a torch attached to a bucket.
- **Inventions falling within Section 20(1) of the Atomic Energy Act, 1962** Examples: Inventions related to compounds of Uranium, Beryllium, Thorium, Plutonium, Radium, Graphite, Lithium, and more as notified by the Central Government.
- **Literary, Dramatic, Musical, Artistic Work**
- Examples: Books, sculptures, drawings, paintings, computer programmes, mathematical calculations, online chatting method, method of teaching, method of learning a language, as they fall under the Copyright Act, 1957.
- **Topography of Integrated Circuits**
- Protection of layout designs of integrated circuits is provided separately under the Semiconductor Integrated Circuit Layout Designs Act, 2000.

Inventions Excluded from Patenting

- **Plants and Animals**
- Plants and animals in whole or in part, including seeds, varieties, and species, along with essentially biological processes for the production or propagation of plants and animals, are excluded from patent protection.
- Examples
 - A new mango variety.
 - A new breed of cow.
 - Seeds of wheat or rice.
 - Natural crossbreeding methods.
- **Traditional Knowledge:** An invention that effectively represents traditional knowledge or is an aggregation or duplication of known properties of traditionally known components is also excluded.

Inventions Excluded from Patenting

- Traditional knowledge refers to the knowledge, practices, and innovations that have been developed, preserved, and passed down from generation to generation within communities, often orally. It usually relates to agriculture, medicine, biodiversity, environment, and culture.
- **☐ Examples of Traditional Knowledge:**
- **Medicine & Healing Practices**
 - Use of **turmeric** for wound healing and as an antiseptic.
 - **Neem** extracts for pest control and medicinal uses.
 - **Ashwagandha** as a stress reliever and immunity booster in Ayurveda.
 - **Unani, Siddha, and Yoga practices.**
- **Agriculture & Food**
 - Traditional methods of **seed preservation** and **crop rotation**.
 - Knowledge of **organic manures** like cow dung and compost.

Patent Infringements

- Once a patent is granted to the applicant, they gain the exclusive right to use or exploit the invention in any capacity. Unauthorized use of the invention without prior permission from the owner constitutes patent infringement. Infringements can be categorized into two types:
- **Direct Infringement:** Occurs when a product is substantially close to any patented product or when the marketing or commercial use of the invention is carried out without the owner's permission.
- **Indirect Infringement**
- Happens when a person or company does not directly make, use, or sell a patented invention, but instead aids, assists, or induces another party to infringe a patent.

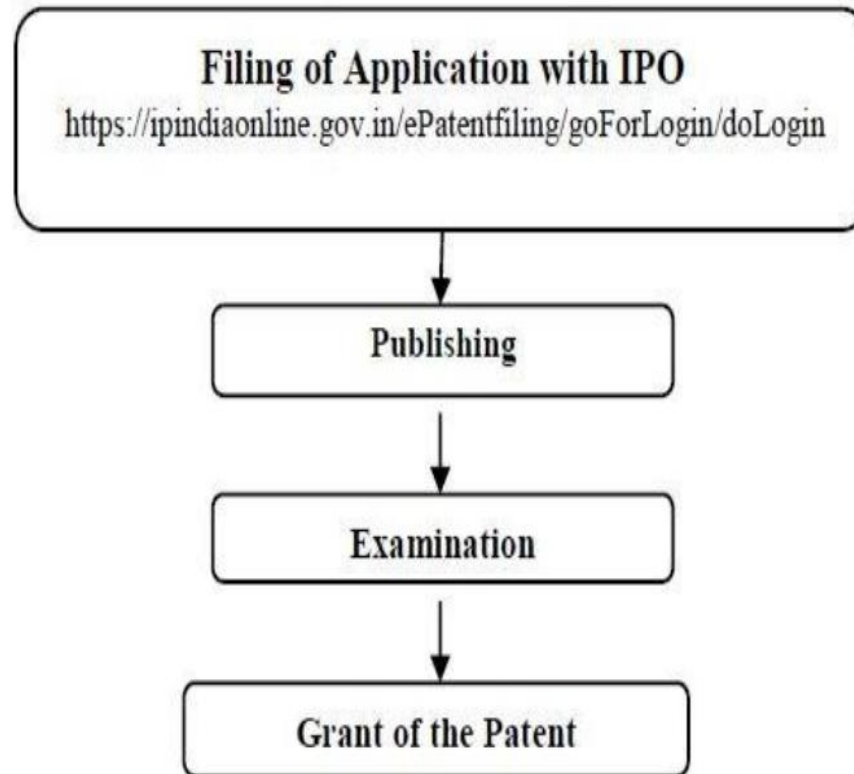
Patent Infringements

- **Types of indirect infringement**
- **Contributory Infringement**
- Supplying a **component, material, or tool** that has no substantial non-infringing use, and is **specially made** for use in a patented invention.
- Example: Selling a unique chip that can only be used inside a patented medical device.
- **Induced Infringement**
- Actively **encouraging, instructing, or persuading** another to infringe.
- Example: Publishing a manual or advertisement that teaches customers how to assemble a patented product without authorization.

Process of Patenting

- In India, the process of granting a patent is a lengthy procedure that may take anywhere from 3-4 years or more. The major steps involved in this process are:
- 1. Prior art search,
- 2. Choice of Application to be Filed,
- 3. Patent Application Forms,
- 4. Jurisdiction of Filing Patent Application,
- 5. Publication,
- 6. Pre-grant Opposition,
- 7. Examination,
- 8. Grant of a Patent,
- 9. Validity of Patent Protection,
- 10. Post-grant Opposition.

Process of Patenting



Process of Patenting

- **Prior Art Search:**
- Before an inventor embarks upon the patent filing process, he has to ensure that his invention is 'novel' as per the criterion for the grant of a patent.
- For this, he has to check whether or not his invention already exists in the public domain.
- For this, he needs to read patent documents and Non-Patent Literature (NPL), scientific journals/reports/magazines, etc.
- The information lying in the public domain in any form either before the filing of the patent application or the priority date of the patent application claiming the invention, is termed as Prior Art.
- The prior art search is carried out on the parameters such as **novelty, patentability, state of the art, infringement, validity, and freedom to operate**. The commonly used databases for prior art search fall into two categories: **Patents Databases and NPL**.

Process of Patenting

- **Patents' Databases :**
- Indian Patent Advanced Search System (InPASS- <http://ipindiaservices.gov.in/publicsearch/>).
- Patentscope (WIPO- <https://www.wipo.int/patentscope/en/>).
- Espacenet (EU- <https://worldwide.espacenet.com/patent/>).
- USPTO (USA- <https://www.uspto.gov/>).
- Google Patents Advanced Search (<https://patents.google.com/advanced>).
- Orbit Intelligence (<https://www.questel.com/business-intelligence-software/orbit-intelligence/>).
- Derwent Innovation (<https://clarivate.com/derwent/solutions/derwent-innovation/>).
- PROQUEST (<https://about.proquest.com/search/?searchKeyword=patent+>).

Process of Patenting

- **Non-Patent Literature (NPL):**
- Scholarly publications: Handbooks, Textbooks, Withdrawn Patents, Encyclopedias, Journals (IEEE, Research Gate, Springer, Wiley Online Library, etc.), Dissertations, NCBI's PubMed, Conference Proceedings, Technical Reports, Public Conferences, etc.
- Industry/trade publications: Industry reviews and public disclosures (Social media, YouTube, Books, Magazines, Datasheets, Blueprints, etc.).
- Others: Newspapers, Websites, Technology blogs, Researchers' websites, etc.

Process of Patenting

- **Choice of Application to be Filed**
- Once a decision has been made to patent the invention, the next step is, what kind of application needs to be filed i.e. provisional patent application or complete (Final) patent application
- Generally, the provisional patent application is preferred for the following reasons:
 - It is cheaper, takes less time, and involves fewer formalities.
 - Any improvements made in the invention after the filing of the provisional application can be included in the final application.
 - In other words, the provisional application does not require complete specifications of the inventions.
 - The application can be filed even though some data is yet to be collected from pending experiments.

Process of Patenting

- A provisional application allows you to secure a priority date for the patent applied.
- However, it is mandatory to file the complete patent application within one year of the filing of the provisional application; otherwise, the application stands rejected.

Process of Patenting

- **Patent Application Forms:**
- As per the Patent Act, 1970 (Section 39) and the Patents Rules, 2003 (Rule 7, 54, 135 and subrule (1) of rule 20, the application for the grant of patent is filed using Form-1 and Form-2.
- The information sought in Form-1 is general in nature i.e. Title of Application, Names of Applicant(s) and Inventor(s), Type of Application (Ordinary, Convention, PCT-NP (PCT- National Phase), Divisional, Patent of Addition, etc.).
- Whereas Form-2 seeks technical information and whether to file the provisional application or complete the application.
- For 'Provisional Application', only 'Description of the Invention' and the 'Abstract' is to be furnished. Whereas, 'Complete Application' requires 'Description of the Invention', 'Abstract', 'Claims' and the manner in which invention has to be performed.

Process of Patenting

- The '**Claims**' of the patent are a very crucial part of the specifications because they define the actual boundary of the invention.
- 'Claims' specify what is actually claimed by the invention and what is being sought to be protected. It clearly describes what the patent does and does not cover.
- The Claims are usually expressed as a declaration of technical particulars articulated in legal terms.
- Claims can be classified into two types
 - a) Independent Claims (stand alone claim) and
 - b) Dependent Claims (dependent on independent claim).
- The Claims must be drafted precisely and carefully in order to seek patent protection and also to protect the invention against potential infringers.

Process of Patenting

India has four offices for filing patent applications(Refer the following table). The applications can be filed only in one of the offices based on the applicant's residence or domicile or place of business or origin of the invention. These are termed as jurisdictions to file patents.

Region/Office	Territorial Jurisdiction
West Region : Patent Office Branch, Mumbai	The States of Maharashtra, Gujarat, Madhya Pradesh, Goa and Chhattisgarh and the Union and the Territories of Daman and Diu & Dadra and Nagar Haveli
South Region :Patent Office Branch, Chennai	The States of Andhra Pradesh, Karnataka, Kerala, Tamil Nadu, Telangana and the Union and the Territories of Pondicherry and Lakshadweep
North Region :Patent Office Branch, New Delhi	The States of Haryana, Himachal Pradesh, Punjab, Rajasthan, Uttar Pradesh, Utarakhand, Delhi and the Union Territory of Chandigarh, Jammu and Kashmir and Ladakh.
Patent Office,Kolkata	The rest of India.

Process of Patenting

- **Publication:**
- Once the patent application has been filed at the Regional Patent Office, the patent application is kept secret for 18 months in the Patent Office.
- After the expiry of 18 months (from the date of filing of the application or the priority claimed date, whichever is earlier), the application is published in the Official Journal of Patent Office (<http://www.ipindia.nic.in/journalpatents.htm>).
- The purpose of publishing the application is to inform the public about the invention.
- The publication of an application is a mandatory step.

Process of Patenting

- **Pre-grant Opposition:**
- If anybody has an objection to the invention claimed in the patent application, he can challenge the application by approaching the Controller of Patents within 6 months from the date of publication. It is termed as Pre-grant Opposition.
- Depending on the outcome of the case, the patent application may be rejected or recommended for the next step, i.e. patent examination.
- Although the patent application is kept secret for 18 months, under special circumstances, this period can be reduced when the patentee(A person/Organization who owns the patent (granted))/applicant plans to sell or license the patent or seek an investor.
- For this, the applicant has to fill a Form-9 and submit it to the Controller General.

Process of Patenting

- **Examination:**
- Patent examination is a critical step in the process of grant of a patent. All the important criteria (novel, inventive step, etc.) are scrutinized by the professionals depending on the content of the invention.
- Usually, the examiner raises certain queries/doubts which need to be addressed by the inventors. Once the examiner is satisfied with the answers received from the inventors, the application is recommended for the grant of a patent.
- It is pertinent to mention that a patent application is not examined automatically after clearing the publication stage.
- The applicant or his representative has to make a request for examination of the patent by filing Form-18A and submitting the same within 48 months from the date of filing of the application.

Process of Patenting

- **Grant of a Patent:**
- After fulfilling all the requirements for the grant of a patent, including all objections/queries raised by the 'Patent Examiner' and the public at large, the patent is granted to the applicant.
- The granted patent is published in the Official Journal of the Patent Office.
- This journal is published every Friday and contains information related to patent applications published under section (u/s) 11A, post-grant publication, restoration of patent, notifications, list of non-working patents and public notices issued by the Patent Office.

Process of Patenting

- **Validity of Patent Protection:**
- The patent protection is granted to an applicant for a limited period, generally 20 years, starting from the date of filing of the application.
- Once a patent is granted for an invention in India, the next vital step is to ensure that it is renewed annually by paying Patent Renewal Fee as per Section 53, Rule 80 of the Indian Patents Act, till the expiry of the patent grant period. Non-payment of Patent Renewal Fee might result in the cancellation of the patent.
- In some countries, patent protection may be extended beyond 20 years. The extension aims to compensate for the time expended on the administrative approval procedure before products can be put on the market.
- The time taken for this procedure means that the patent owner may sometimes not be able to benefit from his right for a considerable period after the grant of the patent.

Process of Patenting

- **Post-grant Opposition :**
- Once the patent has been granted by the Patent Office, it still can be challenged by anyone within one year from the date of publication of the grant of the patent. The granted patent can be challenged either via a Patent Office or in a Court of Law.

Commercialization of a Patent

- The patent owner may grant permission to an individual/ organization / industry to make, use, and sell his patented invention. This takes place according to agreed terms and conditions between the involving parties.
- A patent owner may grant a license to a third party for the reasons mentioned below:
 - The patent owner has a decent job (e.g., university professor) and has no desire or aptitude to exploit the patent on his own.
 - The patent owner may not have the necessary manufacturing facilities.
 - The manufacturing facility is not able to meet the market demand.
 - The patent owner wishes to concentrate on one geographic market; for other geographical markets, he may choose to license the patent rights.

Commercialization of a Patent

- Once the patent is granted, the patentee (person holding the rights to the patent) enjoys the exclusive rights to use the patented invention.
- Only the patentee has the right to license or deal with the patent for any deliberations. Although the validity of the granted patent is for 20 years (from the date of filing a patent application), the patentee is required to furnish information (Form- 27) annually relating to the commercialization/selling of the patent. It is called as 'Working/Licensing of the Patent'.
- The licensing of a patent can be exclusive or non-exclusive.
- In an Exclusive Licence, the patent is sold to only one individual/organization for a fixed time period. During this time period, no other person or entity can exploit the relevant IP except the named licensee.
- In a Non-Exclusive Licence, a patentee can sell his patent rights to as many individuals/parties as he likes.

Commercialization of a Patent

- If the patentee is not able to commercialize his patent within three years from the date of the grant of a patent, any person may submit an application to the Controller of Patents for the grant of Compulsory Licensing (of the patent), subject to the fulfillment of the following conditions:
 - Reasonable requirements of the public concerning the patented invention have not been satisfied.
 - The patented invention is not available to the public at a reasonable price.
 - The patented invention is not worked in the territory of India.

Need for a Patent Attorney/Agent

- In general, applicants can prepare their patent applications and file them without assistance from a patent attorney.
- However, given the complexity of patent documents, it is advisable to seek legal assistance from a patent attorney/agent when drafting a patent application.
- Furthermore, the legislation of many countries requires that an applicant, whose ordinary residence or principal place of business is outside the country, be represented by an attorney or agent qualified in the country (which usually means an agent or attorney who resides and practices in that country).

Can a Worldwide Patent be Obtained

- There is no such term as 'Universal Patent' or 'World Patent' or 'International Patent' as the patent rights are territorial.
- An application for a patent must be filed with a Patent Office of the country in which one wishes to seek patent protection. Unfortunately, this option becomes laborious, cumbersome, timeconsuming and expensive if one wishes to file a patent application in many countries.
- To ease out this issue, many Regional Offices have been established which receive patent applications on behalf of a group of nations e.g. European Patent Office and African Regional Intellectual Property Organization.
- A single application is sufficient to cover many nations that are members of a particular regional office/organization.

Can a Worldwide Patent be Obtained

- However, if one wishes to seek patent protection in several countries worldwide, it is preferred to file an international patent under the Patent Cooperation Treaty (PCT).
- The only condition is that the applicant's country should be a member of PCT. India, along with over 190 nations, is a member of PCT.

Do I Need First to File a Patent in India

- Yes, in general, Indian residents are required to file the patent application first in India. Subsequently, they may file for patent protection in other countries. However, prior approval is needed from the Patent Office. This approval can be waived off under the following circumstances:
- The applicant is not an Indian resident.
- If 6 weeks have expired since the patent application was filed in India by an Indian resident.
- If two or more inventors are working on an invention in a foreign country and one of the inventors is an Indian resident. The invention does not have a potential market in India, and hence the Indian resident seeks Foreign Filing Permission (FFP) from an Indian Patent Office.

Do I Need First to File a Patent in India

- In case of international collaboration, if one part of the invention originated in India and the inventor is an Indian resident, he has to seek permission to file the patent outside India.
- If the invention is related to defense or atomic energy or utility model, the inventor(s) need to seek permission from the Indian Patent Office because inventions related to these domains are not the subject matter of patentability in India.

Patent Related Forms

There are over 30 patent-related forms. Some of Important Patent Application Forms are mentioned below.

Form No.	Title of Form
1	Application for a grant of a patent
2	Provisional/Complete specifications
7	Notice of opposition on grant of a patent
7A	For filing a representation opposing grant of a patent
17	Application for compulsory license
18	Request for examination of the application for patent
21	Request for termination of compulsory license
22	Application for registration of patent agent
27	Statement regarding the working of the patented invention on a commercial scale in India
30	Miscellaneous form to be used when no other form is prescribed

Fee Structure

As per the Patent Act, 1970, and The Patents Rules (1972), the requisite fee has been specified based on the type of form/s to be submitted to the Office (See the Table). Electronically filed applications are 10% cheaper than physical filing.

Item	Natural Person/Startup (Rupees)	Small Entity Alone or with Natural Person/Startup (Rupees)	Others Alone or with Natural Person/Startup/Small Entity (Rupees)
Provisional/Complete Specifications	1,600	4,000	8,000
Request for Early Publication	2,500	6,250	12,500
Request for Examination	4,000	10,000	20,000
Express Request for Examination	5,600	14,000	28,000
Renewal Fees (Annually)			
3rd to 6th Year	800	2,000	4,000
6th to 10th Year	2,400	6,000	12,000
11th to 15th Year	4,800	12,000	24,000
16th to 20th Year	8,000	20,000	40,000

Types of Patent Applications

- **Provisional Application** - A patent application filed when the invention is not fully finalized and some part of the invention is still under experimentation. Such type of application helps to obtain the priority date for the invention.
- **Ordinary Application** - A patent application filed with complete specifications and claims but without claiming any priority date.
- **PCT Application** - An international application filed in accordance with PCT (Patent Corporation Treaty). A single application can be filed to seek patent protection and claim priority in all the member countries of PCT.
- **Divisional Application** - When an application claims more than one invention, the applicant on his own or to meet the official objection on the ground of plurality may divide the application and file two or more applications. This application divided out of the parent one is known as a Divisional Application. The priority date for all the divisional applications will be the same as that of the main (the Parent) Application (Ante-dating).

Types of Patent Applications

- **Patent of Addition Application** - When an invention is a slight modification of the earlier invention for which the patentee has already applied for or has obtained a patent, the applicant can go for 'Patent of Addition', if the modification in the invention is new. Benefit - There is **no need to pay a separate renewal fee for the 'Patent of Addition'**, during the term of the main patent. It expires along with the main patent.
- **Convention Application** - If a patent application has been filed in the Indian Patent Office, and the applicant wishes to file the same invention in the one or more Convention countries (e.g., Paris Convention) by claiming the same priority date on which the application was filed in India, such an application is known as Convention Application. The applicant has to file Convention Application within 12 months from the date of filing in India to claim the same priority date.